

Le Huy Hung

Mechatronics/Robotics Engineering

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SUMMARY

Final-year Mechatronics student at HCMUT, expected to graduate in August 2026 with hands-on experience in robotics programming, embedded systems, control systems, and mechatronic integration. Experienced in developing ROS2 robot systems, automation, computer vision, sensor integration, CAD-based mechanical design, prototype testing, and system-level debugging. Strong in systems thinking, problem-solving, leadership, and cross-functional collaboration.

EDUCATION

Ho Chi Minh City University of Technology (HCMUT)

2022 - Present

Bachelor of Mechatronics Engineering

- Courses: Robotics, Dynamic Systems and Control, Motion Control Systems, Computer Vision, etc.
- TOEIC 820
- C Certification - TEK4VN, Python Certification - fCC

WORK EXPERIENCE

TECHNIXAM

May 2024 - Feb 2025

Intern

- Developed end-to-end mechatronic products involving firmware, mechanical design, and hardware integration.
- Operated, maintained, and troubleshoot production systems.
- Main project: Fully developed a **3D-Printed Manual Sony Camera Lens** and an **Automated PET Filament Recycling Machine**.

Techtronic Industries (TTI) – Milwaukee Tool

Jun 2025 - Sep 2025

New Product Development Engineer Intern

- Applied engineering analysis tools within the NPD Internship Program, including RCA, PFMEA, DFMEA, CPK, Lean, and DFM.
- Supported the development, testing, and issue resolution phases for the **M18 FUEL™ Snow Blower** project.
- Worked cross-functionally to test prototypes, analyze failures, perform root cause analysis, and debug technical issues (reducing from a maximum of 7 NG/day to 2 NG/day).

NovaLabs

May 2025 – Present

Senior Member & Mentor

- Mentored and guided junior members across projects and competitions.
- Oversaw technical quality of client-facing services (**3D Printing, PCBA Design, Custom Mechanical Design**), coordinating cross-functional collaboration between hardware, firmware, and software members.

PERSONAL PROJECT

Autonomous Mobile Robot – Capstone Project

2025 – 2026

Project Lead / Robotics Developer

- Designed mechanical components, 1-layer PCBA, power distribution electrical system, and embedded software architecture for ESP32 and Raspberry Pi 4.
- Performed data analysis, prototype testing, log parsing, and system debugging, improving localization accuracy to within ± 20 mm.
- Implemented ROS2 SLAM, Computer Vision, Gazebo simulation, and a Web UI controller.
- Integrated and fused multi-sensor data (Encoders, IMU, 2D LiDAR, Depth Camera, Proximity), evaluating and optimizing algorithm cycle times to ensure a stable 20 Hz real-time execution loop.

Hexapod Robot

2026

Hardware / Firmware Designer

- Designed 18 DOF structure in SolidWorks and developed 1-layer PCBA and central MCU control.
- Developed embedded firmware in C for ESP32, implementing low-level I2C drivers and a servo control layer for real-time actuation.
- Modeled and implemented Inverse Kinematics and real-time gait generation algorithms, utilizing MATLAB for kinematics simulation.

Other projects: Mini-Line-AGV, Finger-Counter Website, Social-Media Website (Node.js), Robot-Arm Controller...

OTHER EXPERIENCES

MERC 2024 & 2025 – Mechanical Engineering Robot Contest

2024 – 2025

Team Leader / Mechanical Designer

- Coordinated the design of two robots: a miniature model and a pneumatic-driven system.
- Engineered structural frames and calculated specifications for motor torque and pneumatics.
- Mentored team members in embedded programming and electrical circuit design.

NEXCUBE – Modular Electronics Lab Ecosystem

Jan 2026 - Present

Founder / Product Developer

- Managed a team of 5 international members, coordinating task assignments, project timelines, and performance tracking.
- Designed and developed 3D-printed housing, electrical wire bus, and PCBAs for 4 functional prototypes: **Battery Cube, Soldering Cube, Tester Cube, and Power Cube**.
- Researched and developed a magnetic transfer electrical bus contact adapter using pogo-pin interfaces.
- Participated and pitched in APEC Innovation 2026 and Bach Khoa Innovation 2026.

SKILLS

- **Programming:** C/C++, Python, PLC.
- **Mechanical Skills:** 3D Printing (Advanced), SolidWorks, AutoCAD.
- **Robotics & Embedded Systems:** ROS2 (Python), STM32, ESP32.
- **Tools:** Git, Linux, Docker, OpenCV, YOLO, Altium Designer, Proteus, MATLAB/Simulink.